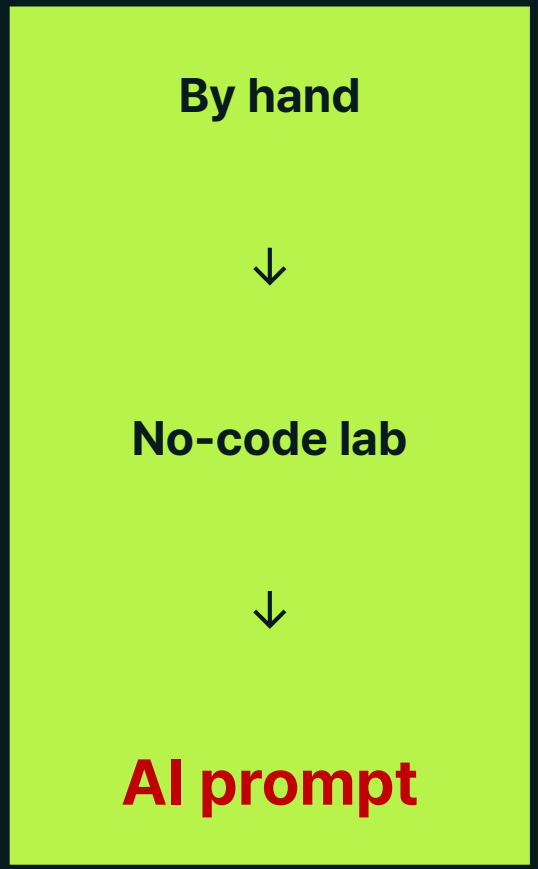
Honest Backtesting Workbook

Step-by-step solutions you can follow with a calculator



Financial ML · Week 10 Workbook

Monthly Sharpe 0.5 is about 1.73 annualized

Monthly excess return 1%, volatility 2%, i.i.d. stable variance

Step 1 · Monthly Sharpe, then annualize

$$SR_{\text{monthly}} = \frac{1\%}{2\%} = 0.5, \qquad SR_{\text{annual}} \approx 0.5 \times \sqrt{12} = 0.5 \times 3.464 \approx 1.73$$

Step 2 · Run 1,000 tests at the 5% level

$1000 \times 0.05 = 50$ significant by chance

Strong autocorrelation breaks simple annualization

The best of 1,000 no-skill strategies has Sharpe about 2.15

3 years, 756 daily obs., N=1,000

Step 1 · Source formula

Approx. best Sharpe by luck $\approx \sqrt{252}\,\dfrac{\sqrt{2\ln N}}{\sqrt{T}}$, N = number of trials, T = number of daily obs

Step 2 · Plug in

$$\sqrt{252}\,\frac{\sqrt{2\ln 1000}}{\sqrt{756}} = 15.87 \times \frac{\sqrt{13.82}}{27.50} = 15.87 \times \frac{3.717}{27.50} \approx 2.15$$

This is neither the DSR nor a strict significance bound

Shading label windows shows which samples to purge

Starts on days 1–10, 3-day return labels, test days 5–7

1. Where

Draw date cells on paper or a notepad.

2. Input

Write each sample's start and 3-day label span.

3. Do and output

Shade test dates; cross out overlapping training samples.

4. Check

Note if endpoints count and if post-test data is allowed.

Prompt for the AI chat

I am a non-major. Tabulate start and end dates of 10 samples with 3-day return labels; the test is days 5–7. Using closed intervals, mark overlapping training samples. Compare walk-forward with purged CV (training on both sides) and explain how future-data use differs from label overlap.

Barriers at twice the volatility sit at ±2%

Start price 100, daily volatility σ = 1%, multiple m = 2, time limit 10 days

Step 1 · Set the upper and lower barriers

$$P_{\text{up}} = 100\,(1 + m\sigma) = 100\,(1 + 2 \times 0.01) = 102, \qquad P_{\text{dn}} = 100\,(1 - m\sigma) = 98$$

Step 2 · Compare closing prices with the barriers in order

Day 0 100 · day 1 100.6 · day 2 101.2 · day 3 101.0 · day 4 101.7 · day 5 102.2

Day 5: 102.2 ≥ 102 → upper barrier first → label +1 (before the 10-day limit)

Judged on closes — intraday highs and lows could touch a barrier earlier

After meta-labeling, precision rises from 40% to 60%

40 wins in 100 primary signals → 30 wins in 50 that pass the meta model (illustrative)

Step 1 · Precision: share of trades taken that worked

$$\text{Precision} = \frac{40}{100} = 0.40 \quad \rightarrow \quad \frac{30}{50} = 0.60$$

Step 2 · Recall: share of the 40 possible wins actually traded

$$\text{Recall} = \frac{40}{40} = 1.00 \quad \rightarrow \quad \frac{30}{40} = 0.75$$

Step 3 · F1: harmonic mean of the two

$$F_1 = \frac{2PR}{P+R}: \quad \frac{2(0.40)(1.00)}{1.40} = 0.57 \quad \rightarrow \quad \frac{2(0.60)(0.75)}{1.35} = 0.67$$

F1 rising from 0.57 to 0.67 means filtering gained more than it missed

With N = 6 and k = 2: 15 splits, 5 paths

CPCV split and path counts — then try N = 10

Step 1 · Ways to choose 2 of 6 groups

$$\binom{6}{2} = \frac{6 \times 5}{2} = 15$$

Step 2 · Paths φ: how often each group is tested

$$\varphi[N, k] = \frac{k}{N}\binom{N}{k} = \frac{2}{6} \times 15 = 5 = \binom{5}{1}$$

Step 3 · With 10 groups

$$N = 10,\ \ k = 2: \quad \binom{10}{2} = 45, \qquad \varphi = \frac{2}{10} \times 45 = 9$$

Larger k gives more paths but a smaller training share (1 – k/N) in each split

Build the triple barrier and CPCV tables by hand

No code — check with a calculator and an AI chat

1. Where

On paper or a spreadsheet, draw cells for closes on days 0–10 and for 6 groups.

2. Input

Write the closes above, σ = 1%, m = 2, N = 6, k = 2.

3. Work and output

Note the first barrier and label; shade the two test groups in each of 15 splits.

4. Check

Label is +1, each group is tested 5 times, and there are 5 paths.

Prompt to paste into an AI chat

I am new to this. Set triple barriers for start price 100, daily volatility 1%, multiple 2 and a 10-day limit, and show in a table which barrier the closes 100, 100.6, 101.2, 101.0, 101.7, 102.2 touch first and the label. Then build tables for CPCV with 6 groups tested 2 at a time: the 15 splits and 5 paths, with each calculation step written out.

CALCULATE BY HAND.

CHECK THE DENOMINATOR.

THEN TRUST THE NUMBER.

Calculate once by hand
and the assumptions show.